

Raksha M

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raksha-portfolio-peach.vercel.app

EDUCATION

The Oxford College Of Engineering, Bangalore <i>Master of Technology in AIML</i>	Bangalore <i>Present (SGPA-9.2)</i>
Government Engineering College, Challakere <i>Bachelor of Engineering in AIML, Gold Medalist</i>	Challakere <i>2021 - 2025 (CGPA-9.31)</i>
Sri Vasavi PU College, Challakere <i>PUC in PCMB - July 2021</i>	Challakere <i>Percentage: 99.83</i>
Warriors' English School, Challakere <i>SSLC - April 2019</i>	Challakere <i>Percentage: 97.28</i>

EXPERIENCE

PSYC Aerospace and Defence Industries Pvt. Ltd. <i>AI Intern - R&D</i>	Oct 2025- Present <i>Bengaluru</i>
- Contributed to R&D of AI/ML solutions for aerospace and defence applications, focusing on computer vision-based detection and segmentation of aerial platforms including drones and aircraft. - Trained and tested YOLOv8 and YOLOv11 models using annotated aerial datasets, performing data annotation, preprocessing, model training, validation, testing, and performance evaluation across various drone and aircraft types. - Developed end-to-end ML workflow pipelines and deployed trained computer vision models into drone-based systems for automated aerial object detection, supporting real-world surveillance and defence-oriented applications.	
Rooman Technologies <i>AI ML Engineer Intern</i>	Sep 2024 – Mar 2025 <i>Bengaluru</i>
- Worked on Machine Learning, Prompt Engineering, Data Analysis, GitHub, and Agile Methodologies. - Explored approaches for visualizing data analysis and machine learning predictions.	

PROJECTS

AgroXAI - Explainable Federated Learning for Plant Disease Detection - Developed a privacy-preserving plant disease detection system using Federated Learning and deep learning, enabling collaborative model training without sharing raw image data. - Integrated Grad-CAM based Explainable AI to generate visual heatmaps highlighting disease-affected regions and improve interpretability of model predictions. - Built a full-stack application using React, Tailwind CSS, Flask, and Python with image prediction, disease visualization, prediction history, and persistent data storage.
Structural Health Monitoring Application - Developed a computer vision-based structural inspection system using YOLO for automated detection and localization of cracks in infrastructure images. - Implemented automated crack analysis and structural health assessment by processing detection results and generating an overall inspection health score. - Built a full-stack application using React, Tailwind CSS, and Flask with image inspection, detection results, dashboard visualization, prediction history, and data storage.
AI Powered Mock Interview Platform - Developed an AI-powered web platform for automated and interactive mock interviews to support technical interview preparation. - Integrated Gemini AI for intelligent interview question generation and AI-driven interactions based on interview context and user responses.

- Built the platform using **Next.js, React, Clerk, Drizzle ORM, and PostgreSQL**, implementing authentication, database management, and responsive user interfaces.

Traffic Deadlock Detection and Prevention Simulator

- Developed a four-way traffic junction simulator demonstrating **deadlock detection and prevention** concepts inspired by operating systems and multi-agent systems.
- Implemented **real-time traffic-state monitoring** across four directions and detected deadlock conditions based on conflicting vehicle states at the junction.
- Designed a signal-prioritization mechanism to resolve detected deadlocks and built the simulator using **React and Flask** for interactive traffic management.

PneumoVision - Pneumonia Detection using Deep Learning

- Developed a deep learning-based medical image classification system for automated pneumonia detection from **chest X-ray images**.
- Implemented and compared **CNN, ResNet18, DenseNet, GoogleNet, and VGG** architectures for pneumonia classification and model performance analysis.
- Evaluated model performance using **precision, recall, confusion matrix, and classification metrics** to analyze prediction effectiveness and errors.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, SQL

AI/ML: Artificial Intelligence, Machine Learning, Deep Learning, Federated Learning, Generative AI, Data Science

Computer Vision: Object Detection, Image Segmentation, YOLOv8, YOLOv11, OpenCV, Grad-CAM

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-learn, Ultralytics, Pandas, NumPy, Matplotlib

Web Technologies: React, Next.js, Tailwind CSS, Flask, Node.js

Developer Tools: VS Code, Jupyter Notebook, Google Colab, Git, GitHub

Core Skills: Data Annotation, Data Preprocessing, Model Training, Model Evaluation, Model Deployment, ML Pipelines

ACHIEVEMENTS

Hack4Sustainability

Nov 2023

Smile Foundation, HackerEarth

Bengaluru

- Led a team during a **30-hour hackathon** focused on developing technology-driven solutions for sustainability and waste management.
- Developed a **Waste-to-Energy Conversion prototype** and Java-based mobile application for waste collection, management, and reward allocation based on waste type.

Mission Amrit Sarovar - Jal Dharohar Sanrakshan Internship

Jul 2022

AICTE, New Delhi

Chitradurga

- Conducted research on **Chitradurga Fort** and the rejuvenation and conservation of water bodies surrounding the fort and hill temples.
- Contributed to the **Jal Dharohar Sanrakshan** initiative under AICTE and received stipends and rewards for the research work.

EXTRA-CURRICULAR ACTIVITIES

Hobbies: Singing, Dancing, Drawing, Painting, Playing Chess

Languages known: Kannada, English, Hindi